Adnan Harun Doğan

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Education

Master of Science in Computer Engineering — *Sep 2022 – Jan 2025*
Middle East Technical University (METU) — *Ankara, Turkey*
- **Thesis:** Incorporating Piecewise Linear Functions with Constant Regions in Backpropagation
- **Advisor:** Prof. Dr. Sinan Kalkan, Co-advisor: Assoc. Prof. Dr. Emre Akbas

Bachelor of Science in Computer Engineering — *Sep 2018 – Jul 2022*
Middle East Technical University (METU) — *Ankara, Turkey*
- **Senior Project:** ReQuest: Automated Question-Answer Generation from Text

Research Experience

Research Intern (Supervisor: Prof. Dr. Christian Holz) — *Apr 2025 – Sep 2025*
Sensing, Interaction, & Perception Laboratory (SIPLab), ETH Zürich — *ETH Zürich, Switzerland*
- Trained Kalman filters to denoise the innovation spectrum under spectral supervision for colored noise
- Built EEG ConvLSTM models for real-time VR cybersickness detection on spectra, +30% over baselines

Research Assistant (Supervisor: Prof. Dr. Sinan Kalkan, Assoc. Prof. Dr. Emre Akbas) — *Feb 2023 – Present*
Image Processing and Pattern Recognition Laboratory (ImageLab), METU — *METU, Ankara, Turkey*
- Contributed to Bucketed Ranking Loss, training DETR/transformer detectors up to 6x faster (ECCV 2024)
- Designed optimal-transport (Sinkhorn-Knopp) and ranking losses for end-to-end DETR-style object detection

Intern (Supervisor: Assoc. Prof. Dr. Erol Sahin, Assoc. Prof. Dr. Bugra Koku) — *Oct 2021 – Feb 2022*
Center for Artificial Intelligence and Robotics (ROMER), METU — *METU, Ankara, Turkey*
- Built an RC-car autonomous-driving stack (CarRL) and trained RL navigation agents in ROS2/Gym/PyTorch
- Approximated multi-agent localization solvers for a Hector-quadrotor drone swarm simulated in ROS/Gazebo

Research Intern (Supervisor: Prof. Dr. Mohammad Abdullah Al-Faruque) — *Jun 2021 – Sep 2021*
Cyber-Physical Systems Laboratory (AICPS), University of California, Irvine — *UC Irvine, California, USA*
- Built multi-modal deep models detecting cardiac and sleep abnormalities from ECG and physiological signals
- Fused time-series and spectral features into challenge-competitive ECG arrhythmia classifiers (CinC 2021)

Selected Publications

Dogan, A. H., Demirel, B. U., & Holz, C. (2027). *Learned-noise Kalman filtering with a body prior for drift-calibrated full-body pose from raw sparse inertial signals.* [Manuscript submitted for publication]†

Dogan, A. H., Arasli, D., Akbas, E., Kalkan, S., & Oksuz, K. (2027). *Variational surrogate Jacobians: A framework for constrained gradient propagation.* [Manuscript submitted for publication]†

Dogan, A. H., Baran, O. U., & Alemdar, H. (2026). *Prediction-driven adaptive mesh refinement with graph neural networks for transonic airfoil flows.* [Manuscript submitted for publication]

Dogan, A. H., Deniz, M., Alemdar, H., & Baran, O. U. (2026). *CoFINN: Conservation flux informed neural networks for physics problems governed by conservation laws.* arXiv. arXiv

Dogan, A. H., Demirel, B. U., & Holz, C. (2026). Frequency-weighted neural Kalman filters. ***IEEE/ICRA'26***. arXiv · PDF · Code

Demirel, B. U., Dogan, A. H., Rossie, J., Möbus, M., & Holz, C. (2025). Beyond subjectivity: Continuous cybersickness detection using EEG-based multitaper spectrum estimation. ***IEEE/TVCG'25***. DOI · arXiv · Code

Yavuz, F., Cam, B. C., Dogan, A. H., Oksuz, K., Kalkan, S., & Akbas, E. (2024). Bucketed ranking-based losses for efficient training of object detectors. ***ECCV'24***. DOI · arXiv · PDF · Code

† Venue withheld while under double-blind review.

Honors and Scholarships

University Entrance Exam: ranked 1466th of 2.27M candidates, OSYM, Turkey — *Aug 2017*

Research Interests and Coursework

Optimization for ML (IE 553, IAM 566, IAM 771): Gradients through Discrete Decisions, Differentiable Optimization, Decision-Focused Learning
Probability & Statistics for ML (IAM 541, STAT 565): Optimal Transport, Neural State Estimation
Deep Learning (CENG 501, CENG 502, CENG 796): Differentiable Ranking and Matching, Conditional Computation